

FASTag Fraud Detection

Complete EDA, Feature Engineering, Machine Learning, Tuning and Evaluation

Project Workflow



Scope: Documentation of the analysis and model outputs produced during the project. Numeric results are taken from the outputs/screenshots supplied during the work.

1. Project Objective

The objective was to build a machine-learning system that classifies FASTag transactions as **Fraud** or **Not Fraud**. The project followed the sequence: data cleaning → EDA → feature engineering → train/test split → Random Forest and XGBoost → stacking → threshold/tuning experiments → final comparison.

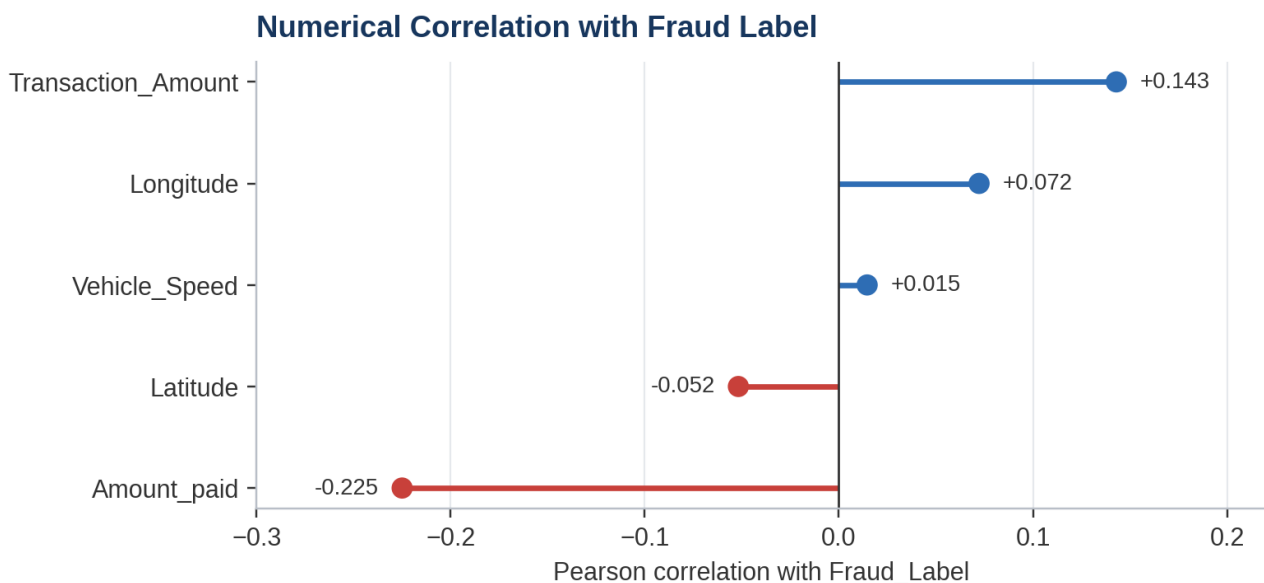
The target variable is **Fraud_Label**: 0 = Not Fraud and 1 = Fraud.

Class	Count	Percentage
Not Fraud (0)	4,017	80.34%
Fraud (1)	983	19.66%
Total	5,000	100.00%

Class imbalance: fraud is the minority class. Therefore, accuracy alone can be misleading. A model can be accurate simply by predicting most transactions as Not Fraud.

2. Data and EDA

Numerical correlation measures linear association between numeric variables and the target. The values observed were:



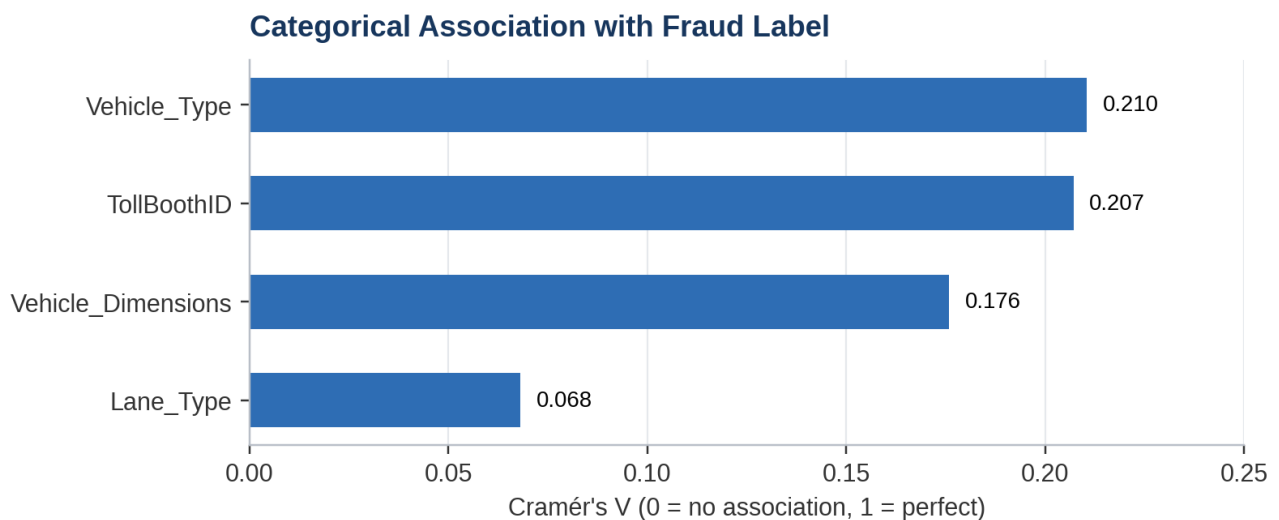
Variable	Correlation
Transaction_Amount	0.142957
Longitude	0.072250
Vehicle_Speed	0.014594
Latitude	-0.051762
Amount_paid	-0.224981

These are weak relationships. This does **not** mean the variables are useless: tree models can learn nonlinear patterns and interactions that Pearson correlation does not capture.

Important modelling decision: Transaction_Amount and Amount_paid were not treated as a simple fraud-correlating pair because payment amount is affected by vehicle/tolling rules. The analysis was kept focused on independent variables and Fraud_Label.

3. Categorical Correlation: Chi-square and Cramér's V

For categorical variables, Pearson correlation is not appropriate. The project used the Chi-square test plus Cramér's V. Chi-square tests whether an association exists; Cramér's V measures its strength from 0 to 1.

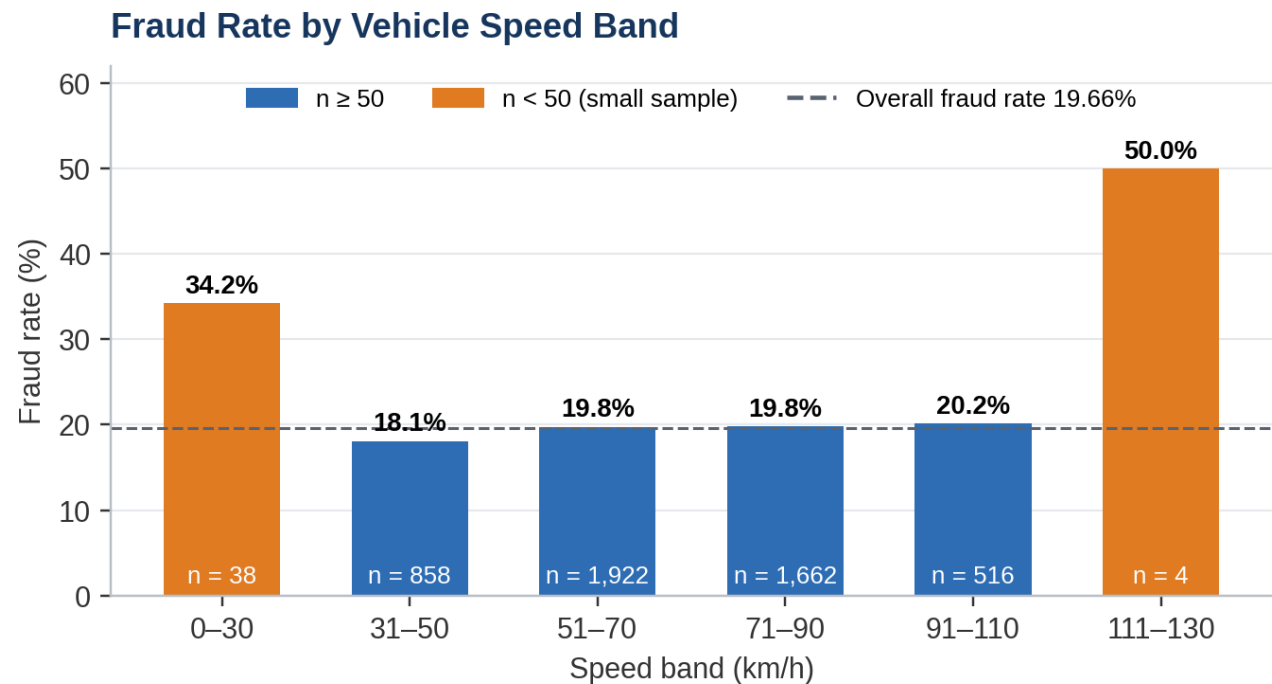


Variable	Chi-square	p-value	Cramér's V
Vehicle_Type	227.367566	2.791048e-46	0.210433
TollBoothID	219.348607	2.048814e-45	0.207070
Vehicle_Dimensions	156.427800	1.076799e-34	0.175760
Lane_Type	24.180581	8.771224e-07	0.068096

The very small p-values indicate statistically detectable association in this sample, while the Cramér's V values are modest. In other words, these variables are related to the fraud label, but none provides a very strong standalone separation.

4. Vehicle Speed EDA

Vehicle speed was examined both as a continuous variable and by speed bands. The observed fraud rates were:



Speed band	Count	Fraud count	Fraud rate
0-30	38	13	34.21%
31-50	858	155	18.07%
51-70	1,922	380	19.77%
71-90	1,662	329	19.80%
91-110	516	104	20.16%
111-130	4	2	50.00%

The 111-130 band has a high percentage but only **four** observations, so it should not be interpreted as strong evidence. This is an example of why counts must be shown with rates.

5. Feature Engineering

Three interaction features were added:

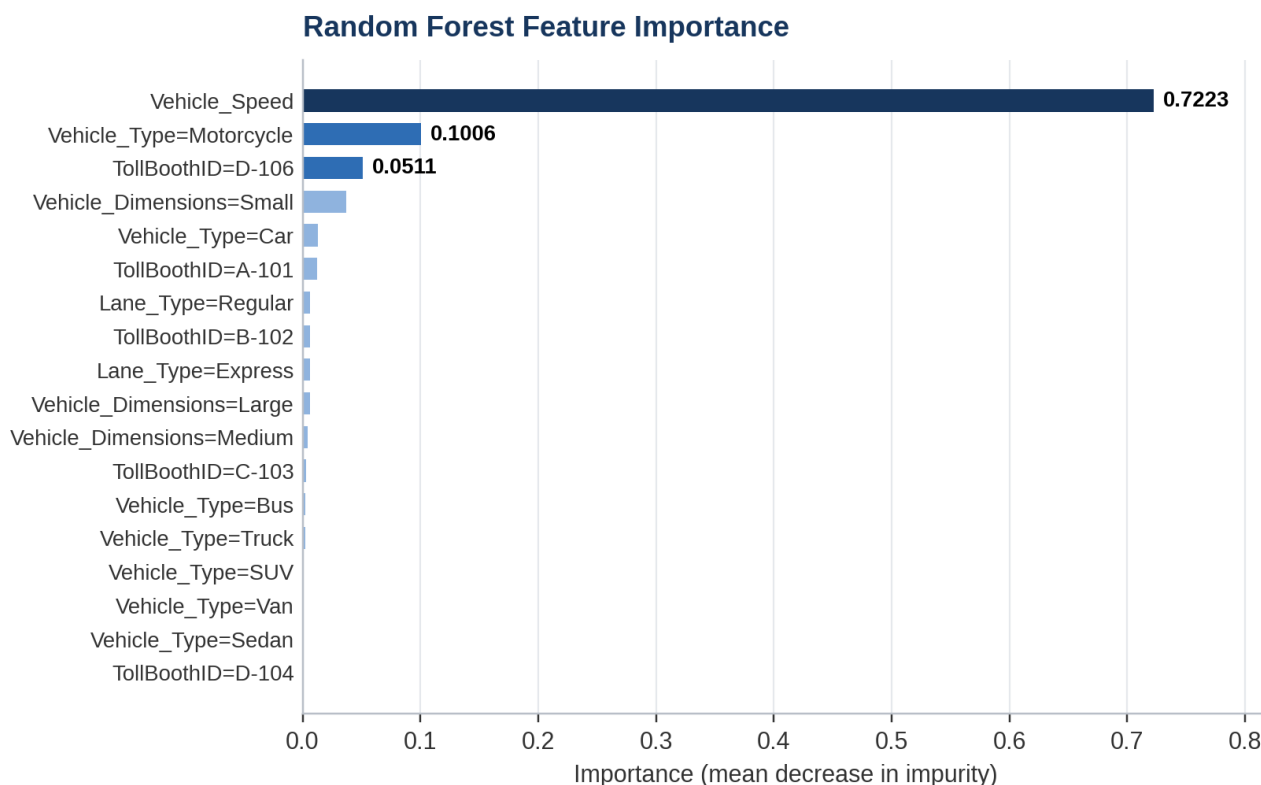
Feature	Purpose
Vehicle Type × Vehicle Dimensions	Captures vehicle category + size combination
Vehicle Type × Lane Type	Captures vehicle category behaviour by lane
Vehicle Dimensions × Lane Type	Captures size behaviour by lane

An interaction can be useful even when each component has weak standalone correlation. Fraud can depend on combinations of characteristics.

Example: an SUV-Large combination is different from a Sedan-Medium combination, and a vehicle using an Express lane can behave differently from the same vehicle category using a Regular lane.

The earlier payment-mismatch analysis was removed because it could confuse the population being measured and blur the distinction between an engineered business rule and the independent-variable-to-fraud analysis.

6. Random Forest Feature Importance



Bars show importance as output by the model; exact values are labelled for the top three features.

The Random Forest output showed Vehicle_Speed importance = **0.722255**, Vehicle_Type=Motorcycle = **0.100553**, and TollBoothID=D-106 = **0.051087**. Feature importance is model-specific: it indicates how much a trained tree ensemble relied on a feature for splitting/prediction. It is not the same as correlation and does not prove causation.

7. Machine Learning Models

Random Forest: an ensemble of many decision trees. Each tree learns different patterns and the ensemble aggregates their predictions. It handles nonlinear relationships and interactions well.

XGBoost: a gradient-boosted tree algorithm. Trees are built sequentially so later trees focus on errors made by earlier trees. It is powerful for structured/tabular data.

Stacking: combines multiple base models. Here, Random Forest and XGBoost generated predictions that were combined by a meta-model.

Train/test split: the model learns on training data and is evaluated on unseen test data. The reported test set contained **1,000** observations: 803 Not Fraud and 197 Fraud.

8. Evaluation Metrics

Metric	Meaning
Accuracy	Percentage of all predictions that are correct.
Precision	Among transactions predicted as Fraud, percentage actually Fraud.
Recall	Among actual Fraud transactions, percentage detected.
F1	Harmonic mean of precision and recall.
ROC-AUC	How well the model ranks positive cases across thresholds using the ROC curve.
PR-AUC	How well the model handles the positive/minority class across thresholds.
TP (True Positive)	Fraud correctly predicted as Fraud.
TN (True Negative)	Not Fraud correctly predicted as Not Fraud.
FP (False Positive)	Not Fraud incorrectly predicted as Fraud.
FN (False Negative)	Fraud incorrectly predicted as Not Fraud.

Fraud context: FN means missed fraud; FP means an unnecessary fraud alert. Both have operational costs, so the appropriate balance depends on the business objective.

9. Initial Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Original Random Forest	.595	.2277	.4416	.3005	.6014	.2618
Original XGBoost	.800	.2857	.0102	.0196	.6053	.2656
Original Stacking	.721	.2153	.1574	.1818	.5886	.2248

Why XGBoost's 80% accuracy was misleading: its confusion matrix was $\begin{bmatrix} 798 & 5 \\ 195 & 2 \end{bmatrix}$. It correctly identified 798 non-fraud cases but caught only 2 of 197 fraud cases. Its fraud recall was therefore just 1.02%.

Random Forest had lower accuracy but caught more fraud: 87 fraud cases in the initial confusion matrix $\begin{bmatrix} 508 & 295 \\ 110 & 87 \end{bmatrix}$. Stacking caught 31 fraud cases with $\begin{bmatrix} 690 & 113 \\ 166 & 31 \end{bmatrix}$.

Original XGBoost	Pred: Not Fraud	Pred: Fraud
Actual: Not Fraud	TN 798	FP 5
Actual: Fraud	FN 195	TP 2

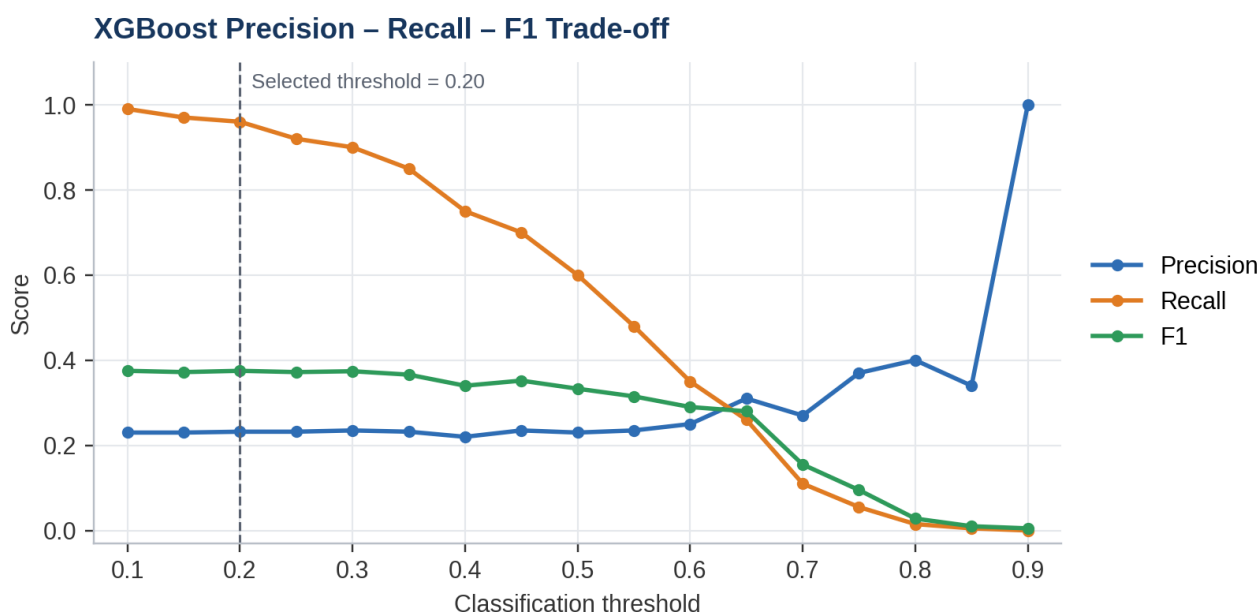
Original Random Forest	Pred: Not Fraud	Pred: Fraud
Actual: Not Fraud	TN 508	FP 295
Actual: Fraud	FN 110	TP 87

10. Feature-Engineered Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Feature-Engineered RF	.586	.2308	.4721	.3100	.6116	.2663
Feature-Engineered XGBoost	.520	.2284	.6041	.3315	.6049	.2611

Feature engineering modestly improved Random Forest: recall rose from 0.4416 to 0.4721 and F1 from 0.3005 to 0.3100. The feature-engineered XGBoost became more fraud-sensitive when class balancing was applied, increasing recall to 0.6041.

11. Classification Threshold Tuning



A classifier usually outputs a probability. The threshold converts that probability into a class. Lowering the threshold generally increases the number of cases labelled Fraud, raising recall but often lowering precision.

A reported XGBoost threshold experiment selected **0.20**. Results: Accuracy = **0.418**, Precision = **0.235940**, Recall = **0.873096**, F1 = **0.371490**.

XGBoost @ threshold 0.20	Pred: Not Fraud	Pred: Fraud
Actual: Not Fraud	TN 246	FP 557
Actual: Fraud	FN 25	TP 172

Thus 172 of 197 fraud transactions were detected, but 557 of 803 non-fraud transactions were incorrectly flagged. This is the precision/recall trade-off in concrete terms.

12. Hyperparameter Tuning + Cross-Validation

Hyperparameters are model settings such as tree depth, number of trees and learning rate. Cross-validation evaluates candidate settings across multiple folds to reduce dependence on one validation split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Original RF	.595	.2277	.4416	.3005	.6014	.2618
FE RF	.586	.2308	.4721	.3100	.6116	.2663
Tuned RF	.440	.2373	.8325	.3694	.6028	.2417
Original XGB	.800	.2857	.0102	.0196	.6053	.2656
FE XGB	.520	.2284	.6041	.3315	.6049	.2611
Tuned XGB	.359	.2312	.9695	.3734	.6052	.2398

Tuning moved the models toward detecting the minority fraud class. Tuned XGBoost reached 96.95% fraud recall, but its accuracy fell to 35.9% and precision was 23.12%. Therefore, tuning did not make the model 'better' on every metric; it changed the operating trade-off.

13. PR-Oriented Optimization

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Optimized Random Forest	.437	.2386	.8477	.3724	.5996	.2410
Optimized XGBoost	.359	.2312	.9695	.3734	.6052	.2398

Optimized Random Forest	Pred: Not Fraud	Pred: Fraud
Actual: Not Fraud	TN 270	FP 533
Actual: Fraud	FN 30	TP 167

Optimized XGBoost	Pred: Not Fraud	Pred: Fraud
Actual: Not Fraud	TN 168	FP 635
Actual: Fraud	FN 6	TP 191

These models are aggressive fraud detectors. The optimized XGBoost missed only 6 of the 197 fraud cases, but it generated 635 false alarms among 803 non-fraud transactions.

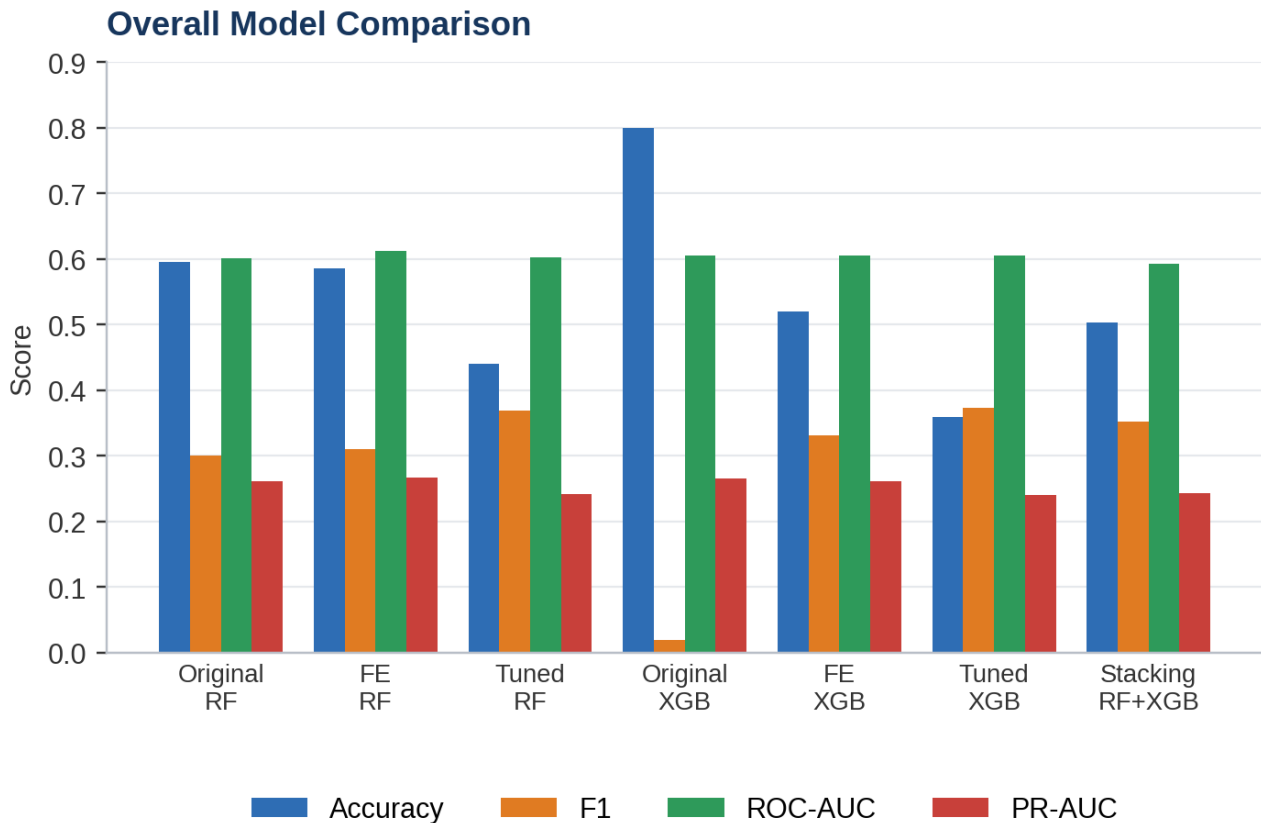
14. Final Stacking RF + XGBoost

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
XGBoost Benchmark	.531	.2354	.6142	.3404	.6090	.2613
Stacking RF + XGBoost	.503	.2368	.6853	.3520	.5934	.2429

Against the feature-engineered XGBoost benchmark, stacking increased precision slightly (0.2354 → 0.2368), recall (0.6142 → 0.6853) and F1 (0.3404 → 0.3520). It reduced accuracy (0.531 → 0.503), ROC-AUC (0.6090 → 0.5934) and PR-AUC (0.2613 → 0.2429).

The correct conclusion is therefore metric-specific: stacking was more aggressive at detecting fraud, but it did not improve every performance measure.

15. Overall Comparison



Model	Accuracy	F1	ROC-AUC	PR-AUC
Original RF	.595	.3005	.6014	.2618
Feature-Engineered RF	.586	.3100	.6116	.2663
Tuned RF	.440	.3694	.6028	.2417
Original XGB	.800	.0196	.6053	.2656
Feature-Engineered XGB	.520	.3315	.6049	.2611
Tuned XGB	.359	.3734	.6052	.2398
Stacking RF+XGB	.503	.3520	.5934	.2429

Central finding: the highest accuracy is not the same as the best fraud detection behaviour. Original XGBoost had 80% accuracy but only 1.02% fraud recall. The tuned models achieved much higher fraud recall by accepting many more false positives.

16. Detailed Interpretation of the Project

Finding 1 — Fraud is not explained by one variable.

Numerical correlations were weak and categorical Cramér's V values were modest. This supports looking for nonlinear and interaction effects.

Finding 2 — Vehicle Type and TollBoothID had the largest categorical associations.

Their Cramér's V values were about 0.21. This means association exists in the sample; it does not establish causation.

Finding 3 — Vehicle Speed was highly important to the Random Forest.

Its feature importance was 0.722255. Importance is model-specific and should not be interpreted as causal evidence.

Finding 4 — Feature engineering helped.

Vehicle Type × Dimensions, Vehicle Type × Lane Type and Vehicle Dimensions × Lane Type were added to represent combinations that individual variables may miss.

Finding 5 — Class balancing and threshold tuning changed the model's behaviour.

Recall increased substantially because the model was encouraged to identify more minority-class cases.

Finding 6 — Stacking changed the operating point.

It improved recall and F1 versus the XGBoost benchmark, while lowering several other metrics.

Finding 7 — Accuracy should not be used alone.

In fraud detection, missing a fraud case and creating a false alert have different consequences. The chosen metric priorities should reflect those costs.

17. Limitations and Next Steps

1. Validate using stratified cross-validation and, where transaction chronology matters, consider time-based validation.
2. Check for data leakage: every production feature must be available at the moment a fraud prediction is made.
3. Choose the classification threshold using business costs for false positives and false negatives.
4. Examine probability calibration if the model's probability output will be used for risk scoring.
5. Use SHAP/permutation importance for explainability and subgroup analysis.
6. Investigate category frequencies and rare groups before interpreting very high fraud rates.
7. Monitor performance over time because fraud patterns can change.

18. Glossary

Term	Simple explanation
EDA	Exploring and understanding data before machine learning.
Independent variable	Input feature used by the model.
Target / dependent variable	Outcome being predicted; Fraud_Label.
Numerical variable	Numeric measurement such as Vehicle_Speed.
Categorical variable	Group/category such as Vehicle_Type.
Correlation	Linear relationship measure for numeric variables.
Chi-square	Test of association between categorical variables.
Cramér's V	Strength of categorical association from 0 to 1.
Feature engineering	Creating new useful variables from existing data.
Interaction feature	Variable representing a combination of two or more inputs.
Random Forest	Ensemble of decision trees.
XGBoost	Sequential gradient-boosted decision trees.
Stacking	Combining multiple model predictions through a meta-model.
Class imbalance	One target class is much more common than another.
Hyperparameter	Model setting chosen during tuning rather than learned as a direct coefficient.
Cross-validation	Repeated train/validation evaluation across folds.
Threshold	Probability cutoff used to decide the predicted class.
Precision	Correctness of positive/fraud alerts.
Recall	Fraction of actual fraud cases detected.
F1	Balance between precision and recall.
ROC-AUC	Threshold-independent ranking measure based on ROC curve.
PR-AUC	Threshold-independent positive-class measure based on precision-recall curve.
False Positive	Non-fraud incorrectly flagged as fraud.
False Negative	Fraud incorrectly classified as non-fraud.

19. Final Conclusion

The FASTag fraud project progressed from EDA to feature engineering and ensemble classification. The EDA showed that fraud is not strongly explained by individual variables, which motivated nonlinear tree-based methods and interaction features.

The experiments also demonstrated why accuracy alone is unsuitable for fraud detection. Original XGBoost reached 80% accuracy but detected only 1.02% of fraud cases. In contrast, tuning and threshold adjustment produced much higher recall, at the cost of many false positives.

The final Stacking RF + XGBoost model achieved 0.503 accuracy, 0.2368 precision, 0.6853 recall, 0.3520 F1, 0.5934 ROC-AUC and 0.2429 PR-AUC on the reported test evaluation. Compared with the feature-engineered XGBoost benchmark, stacking improved recall and F1 but reduced accuracy, ROC-AUC and PR-AUC.

The strongest academic conclusion is: FASTag fraud is better viewed as a multi-feature, nonlinear classification problem rather than a problem driven by one strongly correlated variable. Model evaluation must focus on the fraud class and the business trade-off between missed fraud and false alerts.

Reproducibility note: the numeric results in this document were transcribed from the outputs/screenshots supplied during the project. Small differences may occur if the notebook is rerun with different random seeds, splits, preprocessing or library versions.